INCH-POUND

A-A-60045

13 October 2022

COMMERCIAL ITEM DESCRIPTION

SCREWS, CADMIUM AND HEXAVALENT-CHROMIUM FREE

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1 SCOPE. This commercial item description (CID) covers screws containing no cadmium nor hexavalent-chromium in their finishes.

2 CLASSIFICATION

2.1 Types.

Type I	-	NAS514 - Screw, Machine, 100°, Flat Head, Full Threaded, Alloy Steel
Type II	-	NASM24625 - Screw, Tapping, Thread Cutting, Type BF or BT, Pan Head, Cross-Recessed, Carbon Steel
Type III	-	NASM24629 - Screw, Tapping, Thread Cutting, Types D, F, G, or T, Pan Head, Cruciform Recessed, Carbon Steel, Cadmium Plated
Type IV	-	NASM35190 - Screw, Machine, 82º Flat Countersunk Head, Cruciform Recess, Carbon Steel, Cadmium Plated, Self-Locking and Non-Locking, UNC-2A
Type V	-	NASM35206 - Screw, Machine, Pan Head, Cruciform Recess, Carbon Steel, UNC-2A
Type VI	-	NASM51977 - Setscrew, Hexagon Socket, Half-Dog Point, Alloy Steel, Cadmium Plated, UNC-3A

3 SALIENT CHARACTERISTICS

3.1 Materials. Unless otherwise specified herein, the screws shall be manufactured as specified in the Aerospace Industry Association (AIA) documents listed above by type with the exception of, the plating/finishes shall be free of cadmium and hexavalent-chromium finishes as specified herein. The use of recovered materials made in compliance with regulatory requirements is acceptable provided all requirements of this CID are met.

3.2 Dimensions. Dimensions for all designated types of screws apply after plating/finishing and prior to any application of lubricants (if any). Refer to each individual NAS/NASM specification listed in Section 2.1 for detailed part geometry and tolerance requirements.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 5305

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3.3 Hexavalent-chromium free finish/plating. Carbon and Alloy steel screws – shall be Zinc-Nickel plated in accordance with MIL-PRF-32647, Type K16.

3.4 Interchangeability. When AIA/NASM specifications list interchangeability parts, do not interchange parts containing cadmium or hexavalent-chromium with the cadmium and hexavalent-chromium free parts listed herein without design authority approval.

4 REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation (FAR), or the sustainable acquisition of EO 14057; as applicable.

5 PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

5.3 Contractor certification. The contractor shall certify and maintain substantiating evidence the product offered meets the salient characteristics of this commercial item description and conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to perform or witness any of the inspections it deems necessary to ensure supplies and services conform to prescribed requirements.

6 PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7 NOTES

The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

7.1 Tables. In the following Tables, the CID part numbers are cadmium and hexavalent-chromium free for the designated screw types.

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TABLE I. Type I (-1) Screw, Machine, 100°, Flat Head, Full Threaded, Alloy Steel.

(Use NAS514 for material, heat treatment (if any), and sizes/lengths.)

A-A-60045 Part Number Zinc-Nickel Plated	Reference Part/Dash Number (from NAS514)	A-A-60045 Part Number Zinc-Nickel Plated	Reference Part/Dash Number (from NAS514)
AA60045-1-P440-3	NAS514P440-3	AA60045-1-P632-3	NAS514P632-3
AA60045-1-P440-4	NAS514P440-4	AA60045-1-P632-4	NAS514P632-4
AA60045-1-P440-5	NAS514P440-5	AA60045-1-P632-5	NAS514P632-5
AA60045-1-P440-6	NAS514P440-6	AA60045-1-P632-6	NAS514P632-6
AA60045-1-P440-7	NAS514P440-7	AA60045-1-P632-7	NAS514P632-7
AA60045-1-P440-8	NAS514P440-8	AA60045-1-P632-8	NAS514P632-8
AA60045-1-P440-10	NAS514P440-10	AA60045-1-P632-10	NAS514P632-10
AA60045-1-P440-12	NAS514P440-12	AA60045-1-P632-12	NAS514P632-12
AA60045-1-P440-14	NAS514P440-14	AA60045-1-P632-14	NAS514P632-14
AA60045-1-P440-16	NAS514P440-16	AA60045-1-P632-16	NAS514P632-16
AA60045-1-P440-18	NAS514P440-18	AA60045-1-P632-18	NAS514P632-18
AA60045-1-P440-20	NAS514P440-20	AA60045-1-P632-20	NAS514P632-20
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AA60045-1-P440-24	NAS514P440-24	AA60045-1-P632-24	NAS514P632-24
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---	---	AA60045-1-P632-28	NAS514P632-28
---	---	---	---
---	---	AA60045-1-P632-32	NAS514P632-32
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AA60045-1-P832-4	NAS514P832-4	AA60045-1-P1032-4	NAS514P1032-4
AA60045-1-P832-5	NAS514P832-5	AA60045-1-P1032-5	NAS514P1032-5
AA60045-1-P832-6	NAS514P832-6	AA60045-1-P1032-6	NAS514P1032-6
AA60045-1-P832-7	NAS514P832-7	AA60045-1-P1032-7	NAS514P1032-7
AA60045-1-P832-8	NAS514P832-8	AA60045-1-P1032-8	NAS514P1032-8
AA60045-1-P832-10	NAS514P832-10	AA60045-1-P1032-10	NAS514P1032-10
AA60045-1-P832-12	NAS514P832-12	AA60045-1-P1032-12	NAS514P1032-12
AA60045-1-P832-14	NAS514P832-14	AA60045-1-P1032-14	NAS514P1032-14
AA60045-1-P832-16	NAS514P832-16	AA60045-1-P1032-16	NAS514P1032-16
AA60045-1-P832-18	NAS514P832-18	AA60045-1-P1032-18	NAS514P1032-18
AA60045-1-P832-20	NAS514P832-20	AA60045-1-P1032-20	NAS514P1032-20
---	---	AA60045-1-P1032-22	NAS514P1032-22
AA60045-1-P832-24	NAS514P832-24	AA60045-1-P1032-24	NAS514P1032-24
---	---	AA60045-1-P1032-26	NAS514P1032-26
AA60045-1-P832-28	NAS514P832-28	AA60045-1-P1032-28	NAS514P1032-28
---	---	AA60045-1-P1032-30	NAS514P1032-30
AA60045-1-P832-32	NAS514P832-32	AA60045-1-P1032-32	NAS514P1032-32

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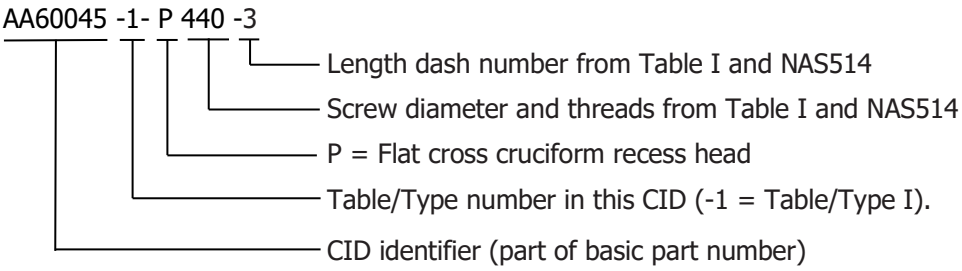
TABLE I. Type I (-1) Screw, Machine, 100°, Flat Head, Full Threaded, Alloy Steel – Continued.

A-A-60045 Part Number Zinc-Nickel Plated	Reference Part/Dash Number (from NAS514)	A-A-60045 Part Number Zinc-Nickel Plated	Reference Part/Dash Number (from NAS514)
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AA60045-1-P428-6	NAS514P428-6	---	---
AA60045-1-P428-7	NAS514P428-7	---	---
AA60045-1-P428-8	NAS514P428-8	AA60045-1-P4524-8	NAS514P4524-8
AA60045-1-P428-10	NAS514P428-10	AA60045-1-P4524-10	NAS514P4524-10
AA60045-1-P428-12	NAS514P428-12	AA60045-1-P4524-12	NAS514P4524-12
AA60045-1-P428-14	NAS514P428-14	AA60045-1-P4524-14	NAS514P4524-14
AA60045-1-P428-16	NAS514P428-16	AA60045-1-P4524-16	NAS514P4524-16
AA60045-1-P428-18	NAS514P428-18	AA60045-1-P4524-18	NAS514P4524-18
AA60045-1-P428-20	NAS514P428-20	AA60045-1-P4524-20	NAS514P4524-20
AA60045-1-P428-22	NAS514P428-22	AA60045-1-P4524-24	NAS514P4524-24
AA60045-1-P428-24	NAS514P428-24	---	---
AA60045-1-P428-26	NAS514P428-26	---	---
AA60045-1-P428-28	NAS514P428-28	AA60045-1-P4524-28	NAS514P4524-28
AA60045-1-P428-30	NAS514P428-30	---	---
AA60045-1-P428-32	NAS514P428-32	AA60045-1-P4524-32	NAS514P4524-32
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---	---		
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AA60045-1-P4624-8	NAS514P4624-8		
AA60045-1-P4624-10	NAS514P4624-10		
AA60045-1-P4624-12	NAS514P4624-12		
AA60045-1-P4624-14	NAS514P4624-14		
AA60045-1-P4624-16	NAS514P4624-16		
AA60045-1-P4624-18	NAS514P4624-18		
AA60045-1-P4624-20	NAS514P4624-20		
AA60045-1-P4624-24	NAS514P4624-24		
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AA60045-1-P4624-28	NAS514P4624-28		
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AA60045-1-P4624-32	NAS514P4624-32		

(Table notes continued on next page)

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PIN construction:



Examples of PIN:

AA60045-1-P440-3 = a NAS514 – Machine screw, 100°, flat cross cruciform recess head, full threaded, 0.1120-40 UNJC-3A threads, 0.188-inch in length, alloy steel, Zinc-Nickel plated.

AA60045-1-P1032-8 = a NAS514 – Machine screw, 100°, flat cross cruciform recess head, full threaded, 0.1900-32 UNJF-3A threads, 0.500-inch in length, alloy steel, Zinc-Nickel plated.

TABLE II. Type II (-2) Screw, tapping, thread cutting, type BF or BT, pan head, cross-recessed, carbon steel.

(Use NASM24625 for material, heat treatment (if any), and sizes/lengths.)

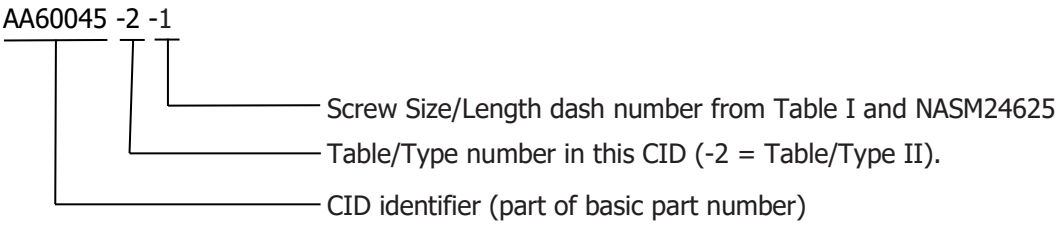
A-A-60045 Part Number Zinc-Nickel Plated	Reference Part/Dash Number (from NASM24625)
AA60045-2-1	MS24625-1
AA60045-2-2	MS24625-2
AA60045-2-3	MS24625-3
AA60045-2-4	MS24625-4
AA60045-2-5	MS24625-5
AA60045-2-6	MS24625-6
AA60045-2-9	MS24625-9
AA60045-2-10	MS24625-10
AA60045-2-11	MS24625-11
AA60045-2-12	MS24625-12
AA60045-2-13	MS24625-13
AA60045-2-14	MS24625-14
AA60045-2-17	MS24625-17
AA60045-2-18	MS24625-18
AA60045-2-19	MS24625-19
AA60045-2-20	MS24625-20
AA60045-2-21	MS24625-21
AA60045-2-22	MS24625-22
AA60045-2-23	MS24625-23
AA60045-2-24	MS24625-24
AA60045-2-25	MS24625-25
AA60045-2-26	MS24625-26
AA60045-2-28	MS24625-28
AA60045-2-29	MS24625-29
AA60045-2-30	MS24625-30
AA60045-2-31	MS24625-31
AA60045-2-32	MS24625-32
AA60045-2-33	MS24625-33
AA60045-2-34	MS24625-34
AA60045-2-35	MS24625-35

A-A-60045 Part Number Zinc-Nickel Plated	Reference Part/Dash Number (from NASM24625)
AA60045-2-36	MS24625-36
AA60045-2-37	MS24625-37
AA60045-2-38	MS24625-38
AA60045-2-39	MS24625-39
AA60045-2-40	MS24625-40
AA60045-2-42	MS24625-42
AA60045-2-43	MS24625-43
AA60045-2-44	MS24625-44
AA60045-2-45	MS24625-45
AA60045-2-46	MS24625-46
AA60045-2-47	MS24625-47
AA60045-2-48	MS24625-48
AA60045-2-49	MS24625-49
AA60045-2-50	MS24625-50
AA60045-2-51	MS24625-51
AA60045-2-52	MS24625-52
AA60045-2-53	MS24625-53
AA60045-2-55	MS24625-55
AA60045-2-56	MS24625-56
AA60045-2-57	MS24625-57
AA60045-2-58	MS24625-58
AA60045-2-59	MS24625-59
AA60045-2-60	MS24625-60
AA60045-2-61	MS24625-61
AA60045-2-62	MS24625-62
AA60045-2-63	MS24625-63
AA60045-2-64	MS24625-64
AA60045-2-65	MS24625-65
AA60045-2-66	MS24625-66

(Table notes continued on next page)

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PIN construction:



Examples of PIN:

AA60045-2-1 = a NASM24625 – Pan head, cross-recessed, tapping, thread cutting screw, type BF or BT, #2 (0.0860 Dia.), carbon steel, 0.188 inch in length, zinc-nickel plated.

AA60045-2-22 = a NASM24625 – Pan head, cross-recessed, tapping, thread cutting screw, type BF or BT, #6 (0.1380 Dia.), carbon steel, 0.750 inch in length, zinc-nickel plated.

AA60045-2-61 = a NASM24625 – Pan head, cross-recessed, tapping, thread cutting screw, type BF or BT, 1/4 (0.2500 Dia.), carbon steel, 1.250 inch in length, zinc-nickel plated.

TABLE III. Type III (-3) Screw, tapping, thread cutting, types D, F, G, or T, pan head, cruciform recessed, carbon steel.

(Use NASM24629 for material, heat treatment (if any), and sizes/lengths.)

A-A-60045 Part Number	Reference (AIA/NASM24629) Dash NO.
AA60045-3-1	MS24629-1
AA60045-3-2	MS24629-2
AA60045-3-3	MS24629-3
AA60045-3-4	MS24629-4
AA60045-3-5	MS24629-5
AA60045-3-6	MS24629-6
AA60045-3-7	MS24629-7
AA60045-3-9	MS24629-9
AA60045-3-10	MS24629-10
AA60045-3-11	MS24629-11
AA60045-3-12	MS24629-12
AA60045-3-13	MS24629-13
AA60045-3-14	MS24629-14
AA60045-3-15	MS24629-15
AA60045-3-16	MS24629-16
AA60045-3-17	MS24629-17
AA60045-3-20	MS24629-20
AA60045-3-21	MS24629-21
AA60045-3-22	MS24629-22
AA60045-3-23	MS24629-23
AA60045-3-24	MS24629-24
AA60045-3-25	MS24629-25
AA60045-3-26	MS24629-26
AA60045-3-27	MS24629-27
AA60045-3-28	MS24629-28
AA60045-3-30	MS24629-30

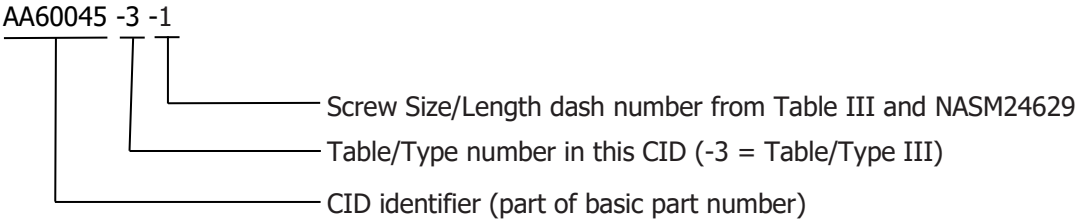
A-A-60045 Part Number	Reference (AIA/NASM24629) Dash NO.
AA60045-3-44	MS24629-44
AA60045-3-45	MS24629-45
AA60045-3-46	MS24629-46
AA60045-3-47	MS24629-47
AA60045-3-48	MS24629-48
AA60045-3-49	MS24629-49
AA60045-3-50	MS24629-50
AA60045-3-51	MS24629-51
AA60045-3-52	MS24629-52
AA60045-3-56	MS24629-56
AA60045-3-57	MS24629-57
AA60045-3-58	MS24629-58
AA60045-3-59	MS24629-59
AA60045-3-60	MS24629-60
AA60045-3-61	MS24629-61
AA60045-3-62	MS24629-62
AA60045-3-63	MS24629-63
AA60045-3-64	MS24629-64
AA60045-3-65	MS24629-65
AA60045-3-66	MS24629-66
AA60045-3-67	MS24629-67
AA60045-3-68	MS24629-68
AA60045-3-69	MS24629-69
AA60045-3-70	MS24629-70
AA60045-3-71	MS24629-71
AA60045-3-72	MS24629-72

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TABLE III. Type III (-3) Screw, tapping, thread cutting, types D, F, G, or T, pan head, cruciform recessed, carbon steel. – Continued.

AA60045-3-33	MS24629-33	AA60045-3-73	MS24629-73
AA60045-3-34	MS24629-34	AA60045-3-74	MS24629-74
AA60045-3-35	MS24629-35	AA60045-3-75	MS24629-75
AA60045-3-36	MS24629-36	AA60045-3-76	MS24629-76
AA60045-3-37	MS24629-37	AA60045-3-77	MS24629-77
AA60045-3-38	MS24629-38	AA60045-3-78	MS24629-78
AA60045-3-39	MS24629-39	AA60045-3-79	MS24629-79
AA60045-3-40	MS24629-40		
AA60045-3-41	MS24629-41		

PIN construction:



Examples of PIN:

AA60045-3-1 = a NASM24629 – Pan head, cruciform recessed, tapping, thread cutting screw, types D, F, G, or T, 0.0860-56, carbon steel, 0.125 inch in length, 0.031 min grip, zinc-nickel plated.

AA60045-3-22 = a NASM24629 – Pan head, cruciform recessed, tapping, thread cutting screw, types D, F, G, or T, 0.1380-32, carbon steel, 0.313 inch in length, 0.141 min grip, zinc-nickel plated.

AA60045-3-70 = a NASM24629 – Pan head, cruciform recessed, tapping, thread cutting screw, types D, F, G, or T, 0.1900-24, carbon steel, 1.000 inch in length, 0.766 min grip, zinc-nickel plated.

TABLE IV. Type IV (-4) Screw, machine, 82º flat countersunk head, cruciform recess, carbon steel, self-locking and non-locking, UNC-2A.

(Use NASM35190 for material, heat treatment (if any), and sizes/lengths.)

A-A-60045 Part Number	Reference Dash NO. (from NASM35190)	A-A-60045 Part Number	Reference Dash NO. (from NASM35190)
AA60045-4-201	MS35190-201	AA60045-4-242	MS35190-242
AA60045-4-202	MS35190-202	AA60045-4-243	MS35190-243
AA60045-4-203	MS35190-203	AA60045-4-244	MS35190-244
AA60045-4-204	MS35190-204	AA60045-4-245	MS35190-245
AA60045-4-205	MS35190-205	AA60045-4-246	MS35190-246
AA60045-4-206	MS35190-206	AA60045-4-247	MS35190-247
AA60045-4-207	MS35190-207	AA60045-4-248	MS35190-248
AA60045-4-208	MS35190-208	AA60045-4-249	MS35190-249
AA60045-4-209	MS35190-209	AA60045-4-250	MS35190-250
AA60045-4-210	MS35190-210	AA60045-4-251	MS35190-251
AA60045-4-211	MS35190-211	AA60045-4-252	MS35190-252
AA60045-4-212	MS35190-212	AA60045-4-253	MS35190-253
AA60045-4-213	MS35190-213	AA60045-4-254	MS35190-254
AA60045-4-214	MS35190-214	AA60045-4-255	MS35190-255
AA60045-4-215	MS35190-215	AA60045-4-256	MS35190-256
AA60045-4-216	MS35190-216	AA60045-4-257	MS35190-257

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TABLE IV. Type IV (-4) Screw, machine, 82º flat countersunk head, cruciform recess, carbon steel, self-locking and non-locking, UNC-2A – Continued.

AA60045-4-217	MS35190-217	AA60045-4-258	MS35190-258
AA60045-4-218	MS35190-218	AA60045-4-259	MS35190-259
AA60045-4-219	MS35190-219	AA60045-4-260	MS35190-260
AA60045-4-220	MS35190-220	AA60045-4-261	MS35190-261
AA60045-4-221	MS35190-221	AA60045-4-262	MS35190-262
AA60045-4-222	MS35190-222	AA60045-4-263	MS35190-263
AA60045-4-223	MS35190-223	AA60045-4-264	MS35190-264
AA60045-4-224	MS35190-224	AA60045-4-265	MS35190-265
AA60045-4-225	MS35190-225	AA60045-4-266	MS35190-266
AA60045-4-226	MS35190-226	AA60045-4-267	MS35190-267
AA60045-4-227	MS35190-227	AA60045-4-268	MS35190-268
AA60045-4-228	MS35190-228	AA60045-4-269	MS35190-269
AA60045-4-229	MS35190-229	AA60045-4-270	MS35190-270
AA60045-4-230	MS35190-230	AA60045-4-271	MS35190-271
AA60045-4-231	MS35190-231	AA60045-4-272	MS35190-272
AA60045-4-232	MS35190-232	AA60045-4-273	MS35190-273
AA60045-4-233	MS35190-233	AA60045-4-274	MS35190-274
AA60045-4-234	MS35190-234	AA60045-4-275	MS35190-275
AA60045-4-235	MS35190-235	AA60045-4-276	MS35190-276
AA60045-4-236	MS35190-236	AA60045-4-277	MS35190-277
AA60045-4-237	MS35190-237	AA60045-4-278	MS35190-278
AA60045-4-238	MS35190-238	AA60045-4-279	MS35190-279
AA60045-4-239	MS35190-239	AA60045-4-280	MS35190-280
AA60045-4-240	MS35190-240	AA60045-4-281	MS35190-281
AA60045-4-241	MS35190-241	AA60045-4-282	MS35190-282
AA60045-4-283	MS35190-283	AA60045-4-324	MS35190-324
AA60045-4-284	MS35190-284	AA60045-4-325	MS35190-325
AA60045-4-285	MS35190-285	AA60045-4-326	MS35190-326
AA60045-4-286	MS35190-286	AA60045-4-327	MS35190-327
AA60045-4-287	MS35190-287	AA60045-4-328	MS35190-328
AA60045-4-288	MS35190-288	AA60045-4-329	MS35190-329
AA60045-4-289	MS35190-289	AA60045-4-330	MS35190-330
AA60045-4-290	MS35190-290	AA60045-4-331	MS35190-331
AA60045-4-291	MS35190-291	AA60045-4-332	MS35190-332
AA60045-4-292	MS35190-292	AA60045-4-333	MS35190-333
AA60045-4-293	MS35190-293	AA60045-4-334	MS35190-334
AA60045-4-294	MS35190-294	AA60045-4-335	MS35190-335
AA60045-4-295	MS35190-295	AA60045-4-336	MS35190-336
AA60045-4-296	MS35190-296	AA60045-4-337	MS35190-337
AA60045-4-297	MS35190-297	AA60045-4-338	MS35190-338
AA60045-4-298	MS35190-298	AA60045-4-339	MS35190-339
AA60045-4-299	MS35190-299	AA60045-4-340	MS35190-340
AA60045-4-300	MS35190-300	AA60045-4-341	MS35190-341
AA60045-4-301	MS35190-301	AA60045-4-342	MS35190-342
AA60045-4-302	MS35190-302	AA60045-4-343	MS35190-343
AA60045-4-303	MS35190-303	AA60045-4-344	MS35190-344
AA60045-4-304	MS35190-304	AA60045-4-345	MS35190-345
AA60045-4-305	MS35190-305	AA60045-4-346	MS35190-346
AA60045-4-306	MS35190-306	AA60045-4-347	MS35190-347
AA60045-4-307	MS35190-307	AA60045-4-348	MS35190-348
AA60045-4-308	MS35190-308	AA60045-4-349	MS35190-349
AA60045-4-309	MS35190-309	AA60045-4-350	MS35190-350

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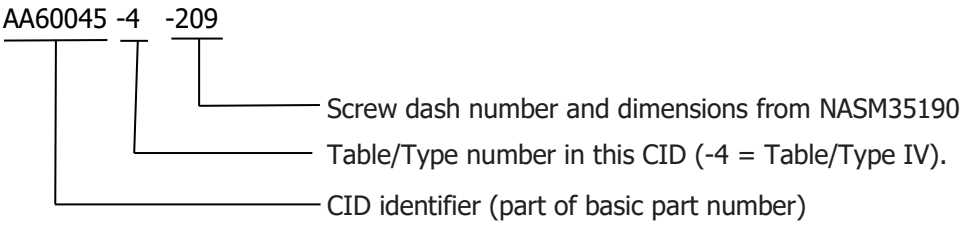
TABLE IV. Type IV (-4) Screw, machine, 82º flat countersunk head, cruciform recess, carbon steel, self-locking and non-locking, UNC-2A – Continued.

AA60045-4-310	MS35190-310
AA60045-4-311	MS35190-311
AA60045-4-312	MS35190-312
AA60045-4-313	MS35190-313
AA60045-4-314	MS35190-314
AA60045-4-315	MS35190-315
AA60045-4-316	MS35190-316
AA60045-4-317	MS35190-317
AA60045-4-318	MS35190-318
AA60045-4-319	MS35190-319
AA60045-4-320	MS35190-320
AA60045-4-321	MS35190-321
AA60045-4-322	MS35190-322
AA60045-4-323	MS35190-323

NOTES:

Add "L" after PIN to designate self-locking thread.

PIN construction:



Examples of PIN:

AA60045-4-209 = a NASM35190 – Machine screw, 82º flat countersunk head, cruciform recess, carbon steel, #2-56, 0.125 inch in length, zinc-nickel plated.

AA60045-4-240 = a NASM35190 – Machine screw, 82º flat countersunk head, cruciform recess, carbon steel, #6-32, 0.750 inch in length, zinc-nickel plated.

AA60045-4-240L = a NASM35190 – Machine screw, 82º flat countersunk head, cruciform recess, carbon steel, #6-32, 0.750 inch in length, zinc-nickel plated, self-locking thread.

TABLE V. Type V (-5) Screw, machine, pan head, cruciform recess, carbon steel, UNC-2A.

(Use NASM35206 for material, heat treatment (if any), and sizes/lengths.)

A-A-60045 Part Number	Reference Dash NO. (from NASM35206)	A-A-60045 Part Number	Reference Dash NO. (from NASM35206)	A-A-60045 Part Number	Reference Dash NO. (from NASM35206)
AA60045-5-201	MS35206-201	AA60045-5-227	MS35206-227	AA60045-5-251	MS35206-251
AA60045-5-202	MS35206-202	AA60045-5-228	MS35206-228	AA60045-5-252	MS35206-252
AA60045-5-203	MS35206-203	AA60045-5-229	MS35206-229	AA60045-5-334	MS35206-334
AA60045-5-204	MS35206-204	AA60045-5-230	MS35206-230	AA60045-5-253	MS35206-253
AA60045-5-205	MS35206-205	AA60045-5-327	MS35206-327	AA60045-5-254	MS35206-254
AA60045-5-206	MS35206-206	AA60045-5-231	MS35206-231	AA60045-5-255	MS35206-255
AA60045-5-207	MS35206-207	AA60045-5-232	MS35206-232	AA60045-5-256	MS35206-256
AA60045-5-320	MS35206-320	AA60045-5-233	MS35206-233	AA60045-5-257	MS35206-257
AA60045-5-208	MS35206-208	AA60045-5-234	MS35206-234	AA60045-5-335	MS35206-335
AA60045-5-209	MS35206-209	AA60045-5-328	MS35206-328	AA60045-5-336	MS35206-336
AA60045-5-210	MS35206-210	AA60045-5-235	MS35206-235	AA60045-5-258	MS35206-258

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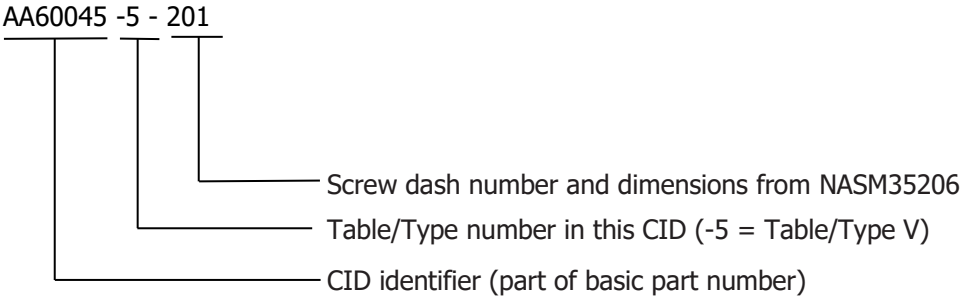
TABLE V. Type V (-5) Screw, machine, pan head, cruciform recess, carbon steel, UNC-2A – Continued.

A-A-60045 Part Number	Reference Dash NO. (from NASM35206)
AA60045-5-321	MS35206-321
AA60045-5-322	MS35206-322
AA60045-5-211	MS35206-211
AA60045-5-212	MS35206-212
AA60045-5-213	MS35206-213
AA60045-5-214	MS35206-214
AA60045-5-215	MS35206-215
AA60045-5-216	MS35206-216
AA60045-5-217	MS35206-217
AA60045-5-323	MS35206-323
AA60045-5-218	MS35206-218
AA60045-5-219	MS35206-219
AA60045-5-220	MS35206-220
AA60045-5-221	MS35206-221
AA60045-5-324	MS35206-324
AA60045-5-222	MS35206-222
AA60045-5-325	MS35206-325
AA60045-5-223	MS35206-223
AA60045-5-326	MS35206-326
AA60045-5-224	MS35206-224
AA60045-5-225	MS35206-225
AA60045-5-226	MS35206-226
AA60045-5-342	MS35206-342
AA60045-5-276	MS35206-276
AA60045-5-277	MS35206-277
AA60045-5-278	MS35206-278
AA60045-5-279	MS35206-279
AA60045-5-343	MS35206-343
AA60045-5-280	MS35206-280
AA60045-5-281	MS35206-281
AA60045-5-282	MS35206-282
AA60045-5-283	MS35206-283
AA60045-5-344	MS35206-344
AA60045-5-284	MS35206-284
AA60045-5-345	MS35206-345
AA60045-5-285	MS35206-285
AA60045-5-286	MS35206-286
AA60045-5-346	MS35206-346
AA60045-5-287	MS35206-287
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AA60045-5-290	MS35206-290
AA60045-5-291	MS35206-291
AA60045-5-347	MS35206-347
AA60045-5-348	MS35206-348
AA60045-5-292	MS35206-292
AA60045-5-293	MS35206-293
AA60045-5-294	MS35206-294
AA60045-5-349	MS35206-349
AA60045-5-295	MS35206-295
AA60045-5-296	MS35206-296
AA60045-5-297	MS35206-297
AA60045-5-298	MS35206-298
AA60045-5-350	MS35206-350
AA60045-5-299	MS35206-299
AA60045-5-329	MS35206-329
AA60045-5-236	MS35206-236
AA60045-5-237	MS35206-237
AA60045-5-361	MS35206-361
AA60045-5-238	MS35206-238
AA60045-5-330	MS35206-330
AA60045-5-362	MS35206-362
AA60045-5-239	MS35206-239
AA60045-5-240	MS35206-240
AA60045-5-241	MS35206-241
AA60045-5-242	MS35206-242
AA60045-5-243	MS35206-243
AA60045-5-244	MS35206-244
AA60045-5-245	MS35206-245
AA60045-5-331	MS35206-331
AA60045-5-246	MS35206-246
AA60045-5-247	MS35206-247
AA60045-5-248	MS35206-248
AA60045-5-249	MS35206-249
AA60045-5-332	MS35206-332
AA60045-5-250	MS35206-250
AA60045-5-333	MS35206-333
AA60045-5-351	MS35206-351
AA60045-5-300	MS35206-300
AA60045-5-301	MS35206-301
AA60045-5-352	MS35206-352
AA60045-5-302	MS35206-302
AA60045-5-303	MS35206-303
AA60045-5-304	MS35206-304
AA60045-5-305	MS35206-305
AA60045-5-306	MS35206-306
AA60045-5-353	MS35206-353
AA60045-5-354	MS35206-354
AA60045-5-307	MS35206-307
AA60045-5-355	MS35206-355
AA60045-5-308	MS35206-308
AA60045-5-309	MS35206-309
AA60045-5-310	MS35206-310
AA60045-5-311	MS35206-311
AA60045-5-356	MS35206-356
AA60045-5-312	MS35206-312
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AA60045-5-358	MS35206-358
AA60045-5-315	MS35206-315
AA60045-5-316	MS35206-316
AA60045-5-317	MS35206-317
AA60045-5-318	MS35206-318
AA60045-5-319	MS35206-319
AA60045-5-359	MS35206-359
AA60045-5-360	MS35206-360
AA60045-5-259	MS35206-259
AA60045-5-260	MS35206-260
AA60045-5-261	MS35206-261
AA60045-5-262	MS35206-262
AA60045-5-263	MS35206-263
AA60045-5-337	MS35206-337
AA60045-5-264	MS35206-264
AA60045-5-265	MS35206-265
AA60045-5-266	MS35206-266
AA60045-5-267	MS35206-267
AA60045-5-338	MS35206-338
AA60045-5-268	MS35206-268
AA60045-5-339	MS35206-339
AA60045-5-269	MS35206-269
AA60045-5-270	MS35206-270
AA60045-5-340	MS35206-340
AA60045-5-271	MS35206-271
AA60045-5-272	MS35206-272
AA60045-5-273	MS35206-273
AA60045-5-274	MS35206-274
AA60045-5-275	MS35206-275
AA60045-5-341	MS35206-341

(Table notes continued on next page)

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PIN construction



Examples of PIN:

AA60045-5-201 = a NASM35206 – Machine screw, pan head, cruciform recess, carbon steel, 0.0860-56, 0.1250 inch in length, zinc-nickel plated.

AA60045-5-300 = a NASM35206 – Machine screw, pan head, cruciform recess, carbon steel, 0.3125-18, 1.5000 inch in length, zinc-nickel plated.

AA60045-5-240 = a NASM35206 – Machine screw, pan head, cruciform recess, carbon steel, 0.1640-32, 0.1880 inch in length, zinc-nickel plated.

TABLE VI. Type VI (-6) Setscrew, Hexagon Socket, Half-Dog Point, Alloy Steel, UNC-3A.

(Use NASM51977 for material, heat treatment (if any), and sizes/lengths.)

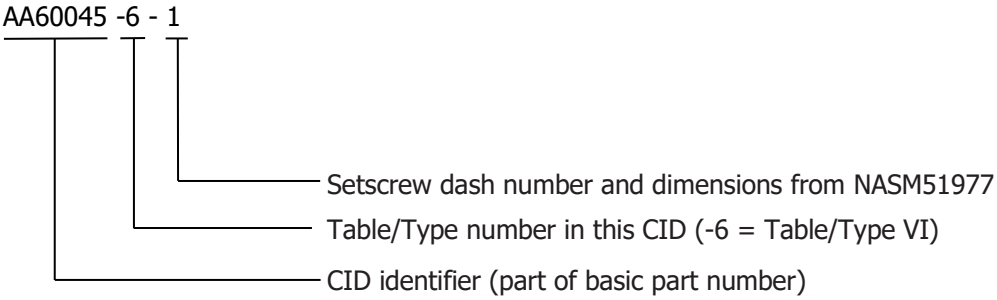
A-A-60045 Part Number	Reference Part NO. (from NASM51977)
AA60045-6-1	MS51977-1
AA60045-6-2	MS51977-2
AA60045-6-3	MS51977-3
AA60045-6-9	MS51977-9
AA60045-6-10	MS51977-10
AA60045-6-11	MS51977-11
AA60045-6-12	MS51977-12
AA60045-6-18	MS51977-18
AA60045-6-19	MS51977-19
AA60045-6-20	MS51977-20
AA60045-6-21	MS51977-21
AA60045-6-22	MS51977-22
AA60045-6-23	MS51977-23
AA60045-6-29	MS51977-29
AA60045-6-30	MS51977-30
AA60045-6-31	MS51977-31
AA60045-6-32	MS51977-32
AA60045-6-33	MS51977-33
AA60045-6-39	MS51977-39
AA60045-6-40	MS51977-40
AA60045-6-41	MS51977-41
AA60045-6-42	MS51977-42
AA60045-6-43	MS51977-43
AA60045-6-49	MS51977-49
AA60045-6-50	MS51977-50
AA60045-6-51	MS51977-51
AA60045-6-52	MS51977-52
AA60045-6-53	MS51977-53
AA60045-6-54	MS51977-54
AA60045-6-55	MS51977-55

A-A-60045 Part Number	Reference Part NO. (from NASM51977)
AA60045-6-61	MS51977-61
AA60045-6-62	MS51977-62
AA60045-6-63	MS51977-63
AA60045-6-64	MS51977-64
AA60045-6-65	MS51977-65
AA60045-6-66	MS51977-66
AA60045-6-67	MS51977-67
AA60045-6-73	MS51977-73
AA60045-6-74	MS51977-74
AA60045-6-75	MS51977-75
AA60045-6-76	MS51977-76
AA60045-6-77	MS51977-77
AA60045-6-78	MS51977-78
AA60045-6-84	MS51977-84
AA60045-6-85	MS51977-85
AA60045-6-86	MS51977-86
AA60045-6-87	MS51977-87
AA60045-6-88	MS51977-88
AA60045-6-94	MS51977-94
AA60045-6-95	MS51977-95
AA60045-6-96	MS51977-96
AA60045-6-97	MS51977-97
AA60045-6-98	MS51977-98

(Table notes continued on next page)

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PIN construction



Examples of PIN:

AA60045-6-1 = a NASM51977 – Setscrew, hexagon socket, half-dog point, alloy steel, 0.0860-56 UNC-3A, 0.125 inch long, zinc-nickel plated.
AA60045-6-1 = a NASM51977 – Setscrew, hexagon socket, half-dog point, alloy steel, 0.0860-56 UNC-3A, 0.125 inch long, zinc-nickel plated.

7.2 Source of documents.

7.2.1 FAR: Copies of the Federal Acquisition Regulations are available online from https://www.acquisition.gov/browse/index/far.

7.2.2 Executive order (EO). Copies of EO 14057 are available online from https://www.fedcenter.gov/_kd/go.cfm?destination=ShowItem&Item_ID=38808.

7.2.3 Military specifications: Copies of military specifications are available online from https://quicksearch.dla.mil.

7.2.4 AIA documents: Copies of Aerospace Industry Association are available online from https://www.aia-aerospace.org/standards/.

7.3 Ordering data. The contract or order should specify the following:

a. Title, number and date of this CID.
b. National stock number, (if available).
c. Material (steel alloy or steel).
d. Hexavalent-chromium and cadmium free part number (PIN) from this CID.
e. Quantity.

7.4 Key words:

a. Alloy steel
b. Carbon steel
c. Plate
d. Plating
e. Steel
f. Zinc-Nickel

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MILITARY INTERESTS:

Custodian:
Army – AT
Navy – AS
Air Force – 71
DLA – CC

Review activities:
Army – AR, AV, CR, MI
Navy – CG, MC, YD
Air Force – 03, 11, 70
DLA – GS4, GS7

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FAS

Preparing Activity:
Army – AT

(Project 5305-2022-015)

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